

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q12-0161

Jack Ramey Wind 138 kV  
August 19, 2013

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0161. The Interconnection Customer proposes a 135.8 MW generating facility interconnecting at the Jack Ramey Wind 138 kV substation in the Virginia area. The request was received on October 02, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 130 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,848,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

The JAK RAMEY WIND - STARKEY 138kV is overloaded to 111.6% of its 250 MVA emergency rating for the loss of the REUSENS - LEESVILLE 345kV line. The project contributes 66.9% DFAX (90.9 MW) to this overload.

The HANCOCK - STARKEY 138kV is overloaded to 110.6% of its 227 MVA emergency rating for the loss of the REUSENS - LEESVILLE 345kV. The project contributes 66.9% DFAX (90.9 MW) to this overload.

The AIBONITO PDS - S CREEK 138kV line is overloaded to 104.1% of its 372 MVA emergency rating for the loss of the CLOVER - BEAR 345kV L. The project contributes 6.1% DFAX (8.2 MW) to this overload.

BEAR - LEAVITT 345kV: 103.5% loading (1,948 MVA) following the outage of the BEAR - LEAVITT 345kV Line 2; DFAX = 15.9%.

For the loss of CLOVER - BEAR 345kV, the CANDELARIA - MONEL 138kV loads to 103.2% of its 388 MVA rating. Project DFAX: 5.1%.

For the loss of BEAR - LEAVITT 345kV Circuit 1, the BEAR - LEAVITT SUBSTATION 345kV Circuit 2 loads to 103.0% of its 1,957 MVA rating. Project DFAX: 15.9%.

BEAR ISLAND - DUNN SIDING 345kV: 102.5% loading (1,570 MVA) following the outage of the LEAVITT - BIRCHWOOD 345kV; DFAX = 14.4%.

For the loss of BEAR - CLOVER 345kV line, the FRANKLIN-ELK GROVE - NEW LONDON SUBSTATION 138kV loads to 101.9% of its 222 MVA rating. Project DFAX: 9.0%.

#### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

#### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

### **4. Required Network Upgrades and Cost Estimates**

| Facility                                         | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| JACK RAMEY WIND - STARKEY 138kV                  | Rebuild line with higher-capacity conductor | \$3,875,000   | 96% (\$3,718,000)     |
| HANCOCK - STARKEY 138kV                          | Rebuild line with higher-capacity conductor | \$4,477,000   | 96% (\$4,295,000)     |
| SOUTH CRK SUBSTATION - AIBONITO PDS 138kV        | Reconductor line                            | \$2,108,000   | 6% (\$127,000)        |
| BEAR - LEAVITT 345kV                             | Reconductor line                            | \$48,639,000  | 4% (\$1,851,000)      |
| MONEL - CANDELARIA 138kV                         | Reconductor line                            | \$12,471,000  | 4% (\$447,000)        |
| LEAVITT - BEAR 345kV Line 2                      | Reconductor line                            | \$58,567,000  | 4% (\$2,228,000)      |
| BEAR ISLAND - DUNN SIDING 345kV L                | Reconductor line                            | \$24,641,000  | 5% (\$1,132,000)      |
| FRANKLIN-ELK GROVE - NEW LONDON SUBSTATION 138kV | Reconductor line                            | \$847,000     | 6% (\$50,000)         |
|                                                  | Total allocated to project                  |               | \$13,848,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.